🛒 Entity Relationship Diagram

RETAIL SALES ANALYSIS — SINGLE TABLE DESIGN

▶ PRIMARY TABLE

RETAIL_SALES

PK	transactions_id	INT
	sale_date	DATE
	sale_time	TIME
IDX	customer_id	INT
	gender	VARCHAR(10)
	age	INT
	category	VARCHAR(20)
	quantiy	INT
	price_per_unit	FLOAT
	cogs	FLOAT
	total_sale	FLOAT

DERIVED VIEW

TIME SHIFTS

- Morning (<12h)
- Afternoon (12-17h)
- Evening (>17h)

DERIVED VIEW

CATEGORIES

- Clothing
- Beauty
- Electronics

DERIVED VIEW

CUSTOMERS

- Unique: ~155
- Gender: M / F
- Age Range: 18-64

SINGLE-TABLE DESIGN · 11 COLUMNS · SELF-CONTAINED ANALYTICS

System Architecture

RETAIL SALES ANALYSIS PROJECT

01 — DATA INPUT LAYER

INPUT

RAW CSV
Retail transactions data imported into MySQL

DATA CLEANING
Remove NULLs, zeros & invalid entries

VALIDATION
Count checks before & after cleaning

02 — PROCESSING LAYER

PROCESS

CRUD QUERIES
SELECT, INSERT, UPDATE, DELETE on retail_sales

AGGREGATIONS
SUM, AVG, COUNT, GROUP BY

WINDOW FN
RANK() OVER PARTITION BY year

DATE/TIME
EXTRACT, HOUR, BETWEEN filters

CTE
WITH hourly_sales AS (...) shift logic

03 — DATABASE LAYER (MYSQL)

DATABASE

PROJECT_1 DB
MySQL database containing retail_sales table

RETAIL_SALES
11 columns · ~2000 rows · Full transaction data

DATE INDEX
sale_date, sale_time for temporal queries

04 — ANALYTICS OUTPUT LAYER

OUTPUT

SALES REPORTS
By category, month, year

CUSTOMER INSIGHTS
Top 5, unique count, avg age

SHIFT ANALYSIS
Morning / Afternoon / Evening orders

BEST MONTH
Highest avg sales per year via RANK()

Data Flow Diagram

RETAIL SALES ANALYSIS — 10 BUSINESS QUESTIONS

FLOW 01 — DATA PIPELINE

CSV File (Raw Sales Data) → IMPORT (MySQL Table Load) → CLEAN (DELETE NULLs & Zeros) → VALIDATE (COUNT(*) Check) → retail_sales (Clean Dataset Ready)

FLOW 02 — DATA EXPLORATION

retail_sales (Source Table) → COUNT(*) (Total Records) → DISTINCT (Unique Customers) → DISTINCT (Product Categories) → Insights (Data Profile)

FLOW 03 — 10 BUSINESS QUERIES

Q1 FILTER
Sales on Specific Date
`WHERE sale_date = '2022-11-05'`

Q2 MULTI-FILTER
Clothing + High Qty + Nov 2022
`WHERE category='Clothing' AND quantiy>=4 AND BETWEEN dates`

Q3 AGGREGATION
Total Sales by Category
`SUM(total_sale) GROUP BY category`

Q4 AGGREGATION
Avg Age — Beauty Customers
`ROUND(AVG(age)) WHERE category='Beauty'`

Q5 FILTER
High-Value Transactions
`WHERE total_sale > 1000`

Q6 GROUP BY
Transactions by Gender & Category
`COUNT(*) GROUP BY category, gender`

Q7 WINDOW FN
Best Selling Month per Year
`RANK() OVER(PARTITION BY year ORDER BY AVG DESC)`

Q8 TOP-N
Top 5 Customers
`SUM(total_sale) GROUP BY customer ORDER BY DESC LIMIT 5`

Q9 COUNT DISTINCT
Unique Customers per Category
`COUNT(DISTINCT customer_id) GROUP BY category`

Q10 CTE + CASE
Orders by Time Shift
`WITH hourly_sales AS (CASE WHEN HOUR < 12 THEN 'Morning'...)`

Schema Overview

RETAIL SALES ANALYSIS — COMPLETE REFERENCE

1
TABLE

11
COLUMNS

10
SQL QUERIES

3
CATEGORIES

3
TIME SHIFTS

retail_sales

DATABASE: PROJECT_1 · ENGINE: MySQL · 11 COLUMNS

KEY	COLUMN NAME	DATA TYPE	CONSTRAINT	DESCRIPTION
PK	transactions_id	INT	NOT NULL	Unique transaction identifier
	sale_date	DATE	NOT NULL	Date of sale (YYYY-MM-DD)
	sale_time	TIME	NOT NULL	Time of sale — used for shift analysis
IDX	customer_id	INT	NOT NULL	Customer reference (logical FK)
	gender	VARCHAR(10)	NOT NULL	Male / Female
	age	INT	NOT NULL	Customer age in years
	category	VARCHAR(20)	NOT NULL	Clothing / Beauty / Electronics
	quantiy	INT	NOT NULL	Units purchased (note: typo in source)
	price_per_unit	FLOAT	NOT NULL	Unit price in USD
	cogs	FLOAT	NOT NULL	Cost of Goods Sold
	total_sale	FLOAT	NOT NULL	Revenue = qty × price_per_unit

CLOTHING
Apparel & accessories — Q2 filter by qty & date

BEAUTY
Beauty products — Q4 avg age analysis

ELECTRONICS
Tech goods — high total_sale values

FILTER QUERIES

```
Q1: WHERE sale_date = '2022-11-05'
Q2: WHERE category + quantiy + BETWEEN dates
Q5: WHERE total_sale > 1000
```

📊 AGGREGATION QUERIES

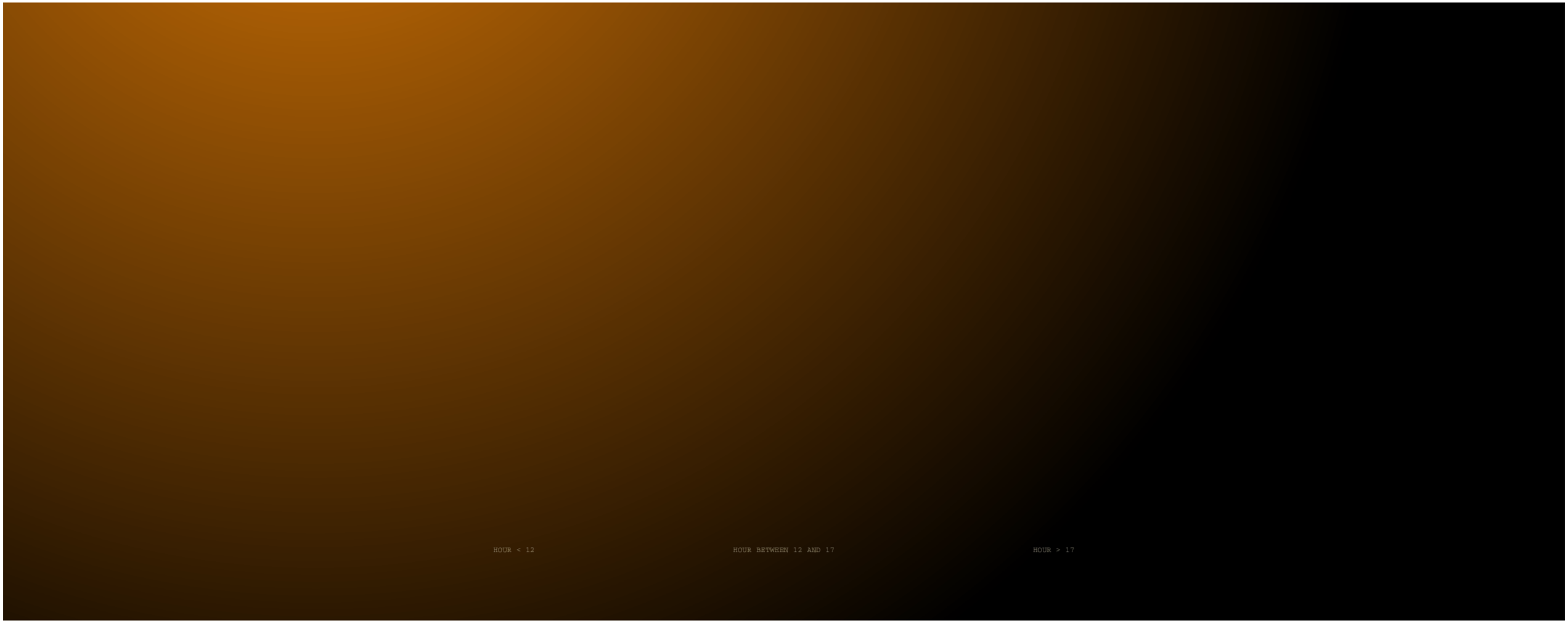
```
Q3: SUM(total_sale) GROUP BY category
Q4: ROUND(AVG(age)) WHERE category='Beauty'
Q6: COUNT(*) GROUP BY category, gender
Q8: SUM ORDER BY DESC LIMIT 5
Q9: COUNT(DISTINCT customer_id)
```

🏆 WINDOW FUNCTION

```
Q7: RANK() OVER(PARTITION BY year)
ORDER BY AVG(total_sale) DESC
→ Best selling month per year
```

⚙️ CTE + CASE

```
Q10: WITH hourly_sales AS (...)
CASE WHEN HOUR(sale_time) < 12 → Morning
BETWEEN 12 AND 17 → Afternoon
ELSE → Evening
```


HOUR < 12
HOUR BETWEEN 12 AND 17
HOUR > 17